Alignment between human reading and large language models varies across word class

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Theories of sentence processing have long posited distinctions in how classes of words are recognized. Bradley (1978), for example, draws a “coarse cut” between the processing of closed-class function words and open-class content words. This theoretical division rests on empirical differences for effects such as word frequency and predictability. Work testing human comprehension with large naturalistic corpora, for example, show qualitative differences between closed- and open-class items in how expectations affect reading times (e.g. Roark et al. 2009) and event-related potentials (ERPs; Brennan & Hale 2019). Oh and Schuler (2023), focusing on how predictability from large-language models (LLMs) of different sizes fit with human reading times, find that this alignment is modulated by certain grammatical features (nouns and adjectives, for example, appear more likely to be under-predicted as models get larger.) The present work builds on this latter effort.

Using the *Natural Stories* reading-time dataset (Futrell et al. 2021) we test the (dis)similarity between surprisal effects on reading times across word class, considering both a “coarse” distinction between closed- and open-class words along with a “finer” distinction between specific parts of speech (POS). We conduct this evaluation across LLMs with a range of parameter sizes to test whether larger models might be better able to capture certain word classes than others. We extend prior efforts by using Bayesian hierarchical modelling which allows us to quantify uncertainty when comparing model fit across grammatical categories. We also include spill-over effects which, while well-documented in the reading literature (Shain 2024), were not discussed in the work mentioned above. These comparisons offer a benchmark for evaluating theories of prediction in reading that incorporate grammatical features.

The dataset includes self-paced reading data from 181 participants who read 1,000 to 10,000 words of English short stories (N = 40 analyzed for this abstract.) We model RTs for open-class (nouns, verbs, adjectives) and closed-class items (determiners, pronouns, adpositions, conjunctions). Predictability via *surprisal* comes from five variants of the pre-trained English OPT LLM spanning 125 million to 6.7 billion parameters (Zhang et al. 2022). The fit between surprisal and RT was evaluated using Bayesian hierarchical linear regression with maximal random effects and fixed effects of length, POS, and surprisal for the current and previous words along with interactions. Model fit was quantified via the expected log pointwise predictive density (ELPD; Vehtari 2017) across LLMs separately for each word category.

Familiarly, higher surprisal increases reading times, but the strength of this effect is modulated by POS (Fig. 1A). For example, verbs show a strong effect of current-word surprisal and an attenuated spillover effect while nouns show the reverse. While all POS show improved fit when surprisal is included relative to a control model, this difference is twice as large for open-class items and determiners compared to other closed-class words (Fig. 1B). Moreover, LLM size modulates goodness-of-fit such that larger models show decreased fit to RTs in a pattern that is strongest for open-class words. Taken together, predictability of open-class words is better captured by LLMs than closed-class words, with declining performance for models with >350 M parameters. This pattern is consistent with theoretical accounts that ground lexical predictions in an inference process that incorporates syntax (e.g. Hale 2014, Eddine et al. 2024, Ohams et al. 2026). We suggest that continued development of such accounts will be aided by considering the asymmetries between word-classes evidenced here.

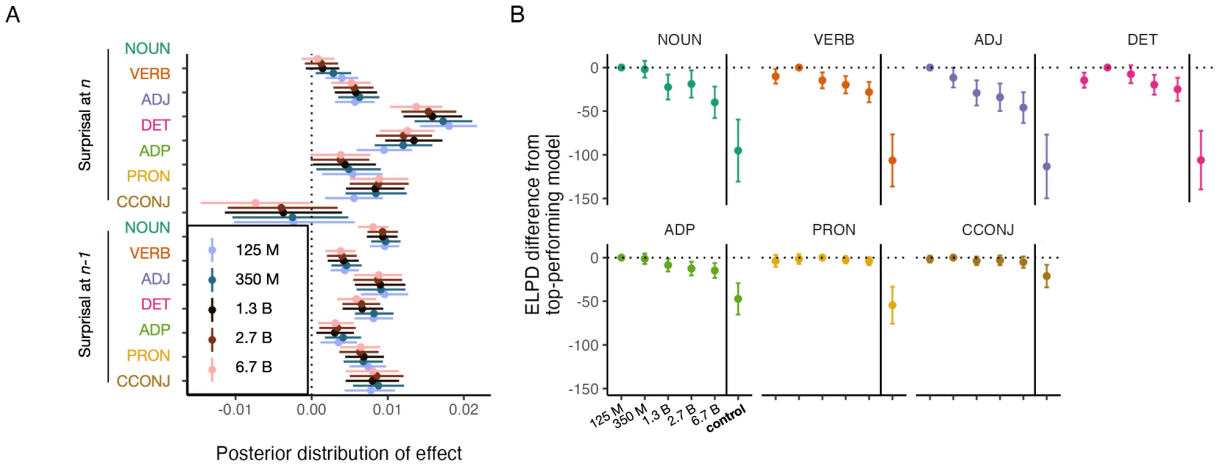

Figure 1: (A) Posterior distributions for coefficients (Median ± 95% CI) estimated from five statistical models fit with *surprisal* from OPT LLMs spanning 125 M up to 6.7 B parameters. Estimates are shown for each part of speech for *surprisal* at both the current word *n* and also at *n-1* to capture spill-over effects. **(B**) ELPD (±2 SE), quantifiers model goodness-of-fit relative to the highest-performing statistical model for each part-of-speech.

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